Q1: Search

- Due 12 Jan at 12:30
- Points 5
- Questions 5
- Available after 31 Oct 2025 at 15:00
- Time limit 60 Minutes
- Allowed attempts 5

This quiz is no longer available as the course has been concluded.

Attempt history

	Attempt	Time	Score
KEPT	Attempt 3	4 minutes	5 out of 5
LATEST	Attempt 3	4 minutes	5 out of 5
	Attempt 2	5 minutes	3.5 out of 5
	Attempt 1	5 minutes	3 out of 5

Correct answers are hidden.

Score for this attempt: 5 out of 5
Submitted 7 Nov 2025 at 21:24
This attempt took 4 minutes.

Question 1
1 / 1 pts
Which statement best describes a successor function used to define a search problem?

- A function which gives the probability of a certain action being successful.
- A function that tells which new state a certain action leads to from a given state.
- A function that tells which states can a given state be reached from.
- A function that lists the costs for taking a certain action.

Question 2
1 / 1 pts

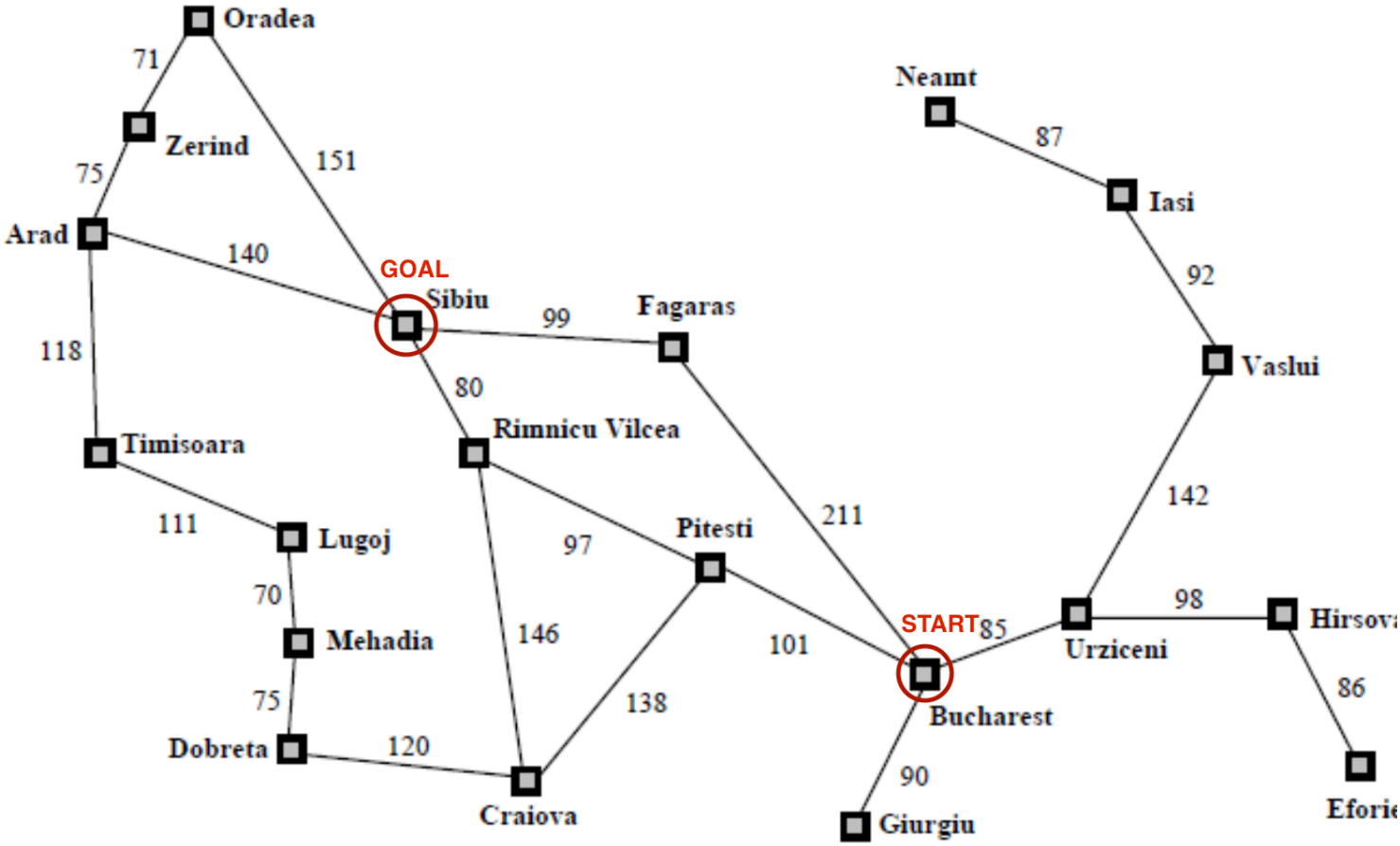
Which statements are true?

- [ ] When you implement DFS, you make use of a queue.
- [x] When you implement DFS, you make use of a stack.
- [ ] When you implement BFS, newly expanded nodes are added to a container and you continue the search from the node which was added most recently to the container.
- [x] When you implement BFS, newly expanded nodes are added to a container and you continue the search from the node which was added first to the container.

Question 3

1 / 1 pts

Consider the following map.



What solution would you find if you use BFS to solve the route planning problem and start in Bucharest and want to find a path to Sibiu?

- [ ] Bucharest - Pitesti - Rimnicu Vilcea - Sibiu
- [x] Bucharest - Fagaras - Sibiu

No solution will be found because BFS will get stuch in a infinite loop going back and forth between Fagaras and Bucharest.

Question 4

1 / 1 pts

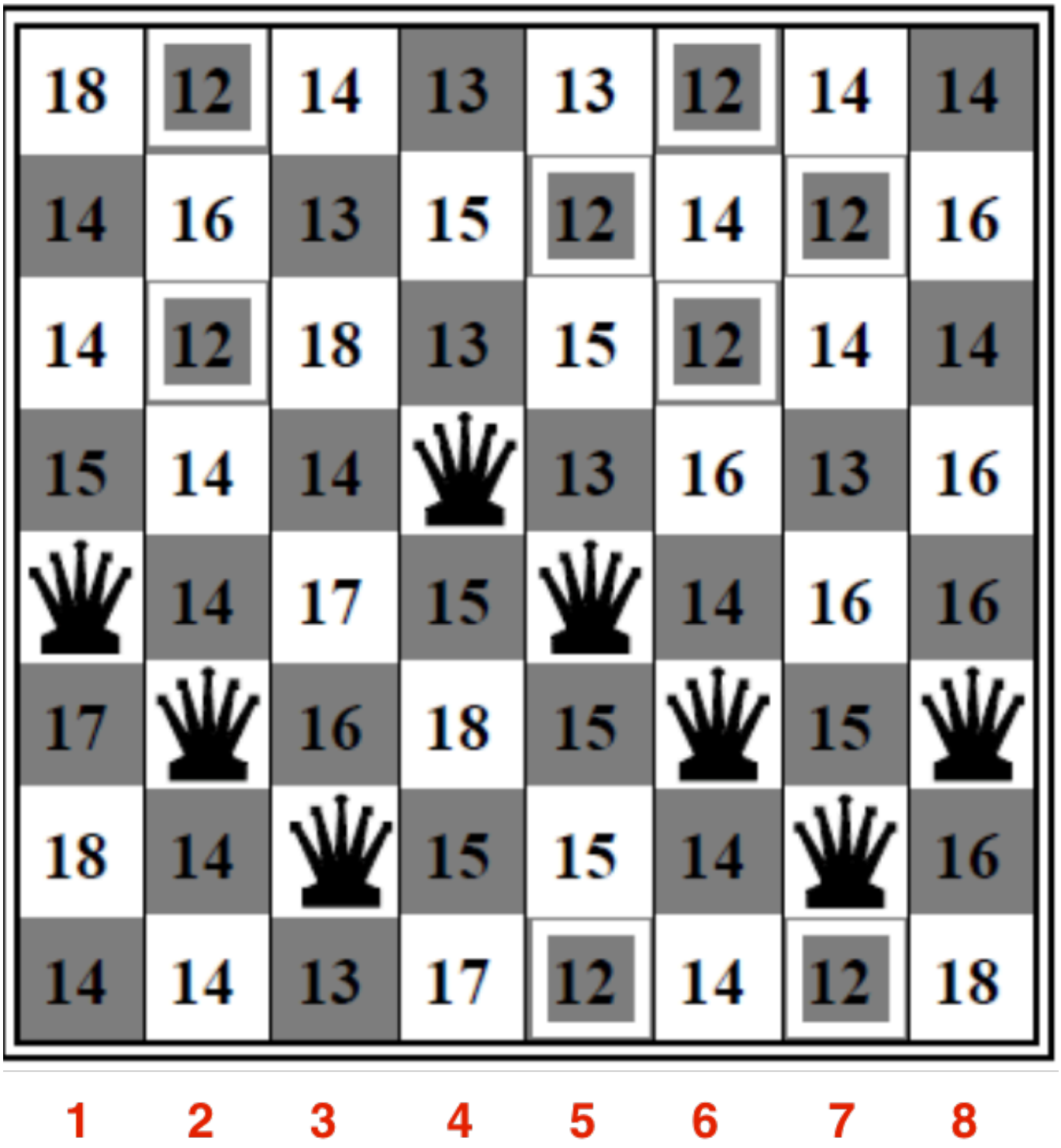
Which description best matches Greedy Best First Search?

- When expanding nodes pick the one with the one that the heuristic estimates has the lowest cost to the goal.
- When expanding nodes pick the one with the lowest accumulated cost to the node.
- When expanding nodes pick the one with the lowest sum of the accumulated cost to the node and the cost that the heuristic estimates to the goal.

Question 5

1 / 1 pts

Consider the game of 8-queens. We want to solve it using hill climbing search. We use the number of pairs of queens under attack as the heuristic. The figure below shows the initial state in the search.



Which queen should we move if we want the best heuristic value after the move (we name the queens after their column, 1,2,...,8)?

- No move is possible since $h > 8$ for all positions.
- Queen 1, 3, 4 or 8
- Queen 2, 5, 6 or 7

Quiz score: 5 out of 5